

Inferring Strong Gravitational Lens Parameters from Single Noisy Images

An amortized BayesFlow posterior estimator, trained on 50,000 lenstronomy simulations per model, evaluated for accuracy, calibration, information gain, and its handling of the mass-sheet degeneracy.

hardware NVIDIA RTX 3060 (12 GB)

dataset 50,000 / 2,000 / 300

training ~2 h / model

peak SNR median 16 (target 10–30)

inference milliseconds / lens

SUMMARY

We train a neural network once on simulated images of strong gravitational lenses, and afterwards it can read off the full Bayesian posterior over 8–9 physical parameters for any new lens image in milliseconds — no per-image fitting, no MCMC. With the source brightness tuned to a realistic peak signal-to-noise ratio (median 16) and the training set raised to 50,000 simulations, the network recovers the Einstein radius, source position, and source size essentially perfectly ($r = 0.999$, $R^2 = 1.00$), the lens ellipticity accurately ($R^2 = 0.92\text{--}0.94$), and even the very weak external shear well (0.80–0.86). Simulation-based calibration finds honest uncertainties for ellipticity and shear and mildly *conservative* (slightly too wide, never overconfident) intervals for the best-measured parameters — a defect we show is a network-capacity effect, not a data effect.

As a stress test, we add an external convergence κ — a parameter that lensing theory says cannot be measured from image shapes alone (the mass-sheet degeneracy). The network correctly reports this: the κ posterior barely contracts from the prior (+16%), shows a -0.96 anti-correlation with the Einstein radius exactly along the theoretical degeneracy direction, and degrades θ_E recovery from $R^2 = 1.00$ to 0.94 while leaving every other parameter untouched.

1 Background: what is being measured

When a massive foreground galaxy sits almost exactly between us and a distant background galaxy, its gravity bends the background light into arcs, multiple images, or a complete Einstein ring. The shape of that distorted image is not random — it encodes the physics of the lens:

- the ring radius traces the total mass enclosed by the lens, summarized by the **Einstein radius θ_E** ;
- the arc shapes trace how flattened the lens mass distribution is (**ellipticity e_1, e_2**);
- subtle extra distortions trace the tidal pull of neighbouring structures (**external shear γ_1, γ_2**);
- the arc configuration and thickness trace where the background source sits (**α_s, β_s**) and how big it is (**R_s**).

The classical way to extract these numbers is to fit a lens model to each image with likelihood-based sampling (MCMC) — accurate, but often hours per system, repeated from scratch for every new image. Upcoming surveys (Euclid, LSST) will find $10^4\text{--}10^5$ new lenses, so per-system MCMC does not scale. Amortized simulation-based inference inverts the cost structure: pay once to train a network on simulated (parameters, image) pairs, then get the posterior $p(\theta \mid \text{image})$ for any new image in a single forward pass.

Two questions structure the analysis:

- How well is each parameter recovered from one noisy image, and are the stated uncertainties statistically calibrated?
- When the external convergence κ is added, does the inference correctly reflect the mass-sheet degeneracy — i.e. report that κ is unidentifiable rather than hallucinate a constraint?

2 The pipeline at a glance

The project is a three-stage command-line pipeline (`main.py`), run once per model variant:

Stage	Command	What it does	Cost
Generate	<code>python main.py generate</code>	Draws parameters from the priors, renders each lens image with lenstronomy, adds noise, saves 52,300 (image, parameters) pairs to <code>data/processed/</code>	~1 min, 15 CPU workers
Train	<code>python main.py train</code>	Trains the BayesFlow approximator (CNN summary network + conditional normalizing flow) on the 50,000-pair training split, with early stopping armed	~2 h, GPU
Evaluate	<code>python main.py evaluate</code>	Draws 2,000 posterior samples for each of 300 held-out systems and produces the recovery, calibration, contraction, posterior-predictive, and fiducial-inference diagnostics below	seconds

The whole cycle was run twice overnight on 18 July 2026: **Run 6** with the 8-parameter model, and **Run 7** with a 9th parameter, the external convergence κ (enabled via `SLI_INCLUDE_KAPPA=1`).

3 Forward model & simulated data

3.1 Lens and source model

Each simulated system is a singular isothermal ellipsoid (SIE) lens with external shear, magnifying a Sérsic-profile source galaxy. The inferred parameter vector is

$$\theta = (\theta_E, e_1, e_2, \gamma_1, \gamma_2, x_s, y_s, R_s) \text{ [+ } \kappa \text{ in Run 7]}$$

Nuisance quantities are fixed and assumed known, per the assignment specification: the lens centroid (measurable from the lens light), the source amplitude, the source Sérsic index ($n = 2$), and the source-light ellipticity.

3.2 Priors

Parameters are drawn from independent uniform priors. Ellipticity is sampled as an axis ratio q and position angle φ , then converted to (e_1, e_2) ; shear is sampled as a modulus and angle, then converted to (γ_1, γ_2) — guaranteeing every draw is physically valid, and producing the ring-shaped joint supports visible in the prior pairplot of the appendix figures.

Parameter	Range	Meaning
θ_E	(0.7, 1.6)''	Einstein radius
q	(0.6, 1.0)	lens axis ratio $\rightarrow (e_1, e_2)$
φ	(0, π)	lens position angle
γ_{ext}	(0, 0.08)	external shear modulus $\rightarrow (\gamma_1, \gamma_2)$
φ_{ext}	(0, π)	external shear angle
x_s, y_s	(−0.3, 0.3)''	source position
R_s	(0.05, 0.25)''	source half-light radius
κ	(0, 0.20)	external convergence — Run 7 only

Table 1. Prior ranges. Lengths in arcseconds, angles in radians.

3.3 Image rendering, noise, and SNR tuning

Images are rendered on a 64×64 grid at 0.05"/pixel (a 3.2" field of view), convolved with a Gaussian PSF of FWHM 0.1", then degraded with Gaussian background noise ($\sigma_{\text{bkg}} = 0.1$ per pixel) plus Poisson shot noise for a 100 s effective exposure, using lenstronomy’s noise utilities.

The assignment targets a realistic peak signal-to-noise ratio of 10–30. The original source amplitude (20) turned out far too faint — median peak SNR ≈ 4 , arcs buried in noise. A sweep over 60 prior draws located the right amplitude:

Source amplitude	min SNR	median SNR	max SNR	Verdict
20 (original)	1.0	3.8	5.5	far below target
60	2.7	8.9	12.1	still short
100	4.1	12.6	16.7	low end of target
150 (adopted)	5.7	16.3	21.3	in target range
200	7.1	19.4	25.1	also acceptable

Table 2. `SOURCE_AMP` tuning sweep (peak SNR over 60 prior draws). The adopted value 150 puts typical images squarely in the 10–30 target.

The realized SNR of the final 50,000-sample datasets confirms the tuning: Run 6 measured min 4.6 / median 16.1 / max 27.4; Run 7 measured min 2.3 / median 16.6 / max 27.5.

The precomputed dataset comprises **50,000 training, 2,000 validation, and 300 held-out test systems** per variant — the upper half of the assignment’s 10^4 – 10^5 budget, reached in steps (8k $\rightarrow$ 20k $\rightarrow$ 50k; see Section 10). Because each simulation is independent, generation parallelizes almost perfectly: the full dataset takes about a minute on 15 CPU workers.

4 Inference method: BayesFlow

We use neural posterior estimation as implemented in BayesFlow: an amortized approximation $q_{\phi}(\theta \mid x)$ to the posterior is trained by maximizing the log-density of the true parameters under the model, across simulated pairs $(\theta, x) \sim p(\theta) p(x \mid \theta)$. No likelihood is ever evaluated — only the simulator is needed. Two networks are learned jointly:

- **Summary network.** A compact CNN compresses each $64 \times 64 \times 1$ image into a 48-dimensional learned summary vector: three convolutional blocks (two 3×3 convolutions each, with 32, 64, 128 filters, followed by max-pooling and batch normalization), global average pooling, and two dense layers. BayesFlow’s built-in summary networks target sets and time series, so this custom CNN is supplied for image data.
- **Inference network.** A conditional coupling-flow normalizing flow (depth 4) maps a standard Gaussian to the posterior over θ , conditioned on the image summary. Sampling the posterior for a new image is a single forward pass.

The target parameters are standardized (z-scored) before entering the flow, since they span very different scales ($\theta_E \sim 1$ vs. $\gamma_i \sim 10^{-2}$).

5 Training

The approximator is trained offline for 40 epochs, batch size 32, Adam optimizer at learning rate 10^{-3} , with **early stopping** on the validation loss (patience 10, best weights restored) as a safeguard against the overfitting seen in the early 8,000-sample pilot. On the RTX 3060 each model trains in ~123 minutes at ~118 ms/step (the same step cost ~500 ms on CPU — the 4–5× GPU speedup is what made a 50k dataset practical).

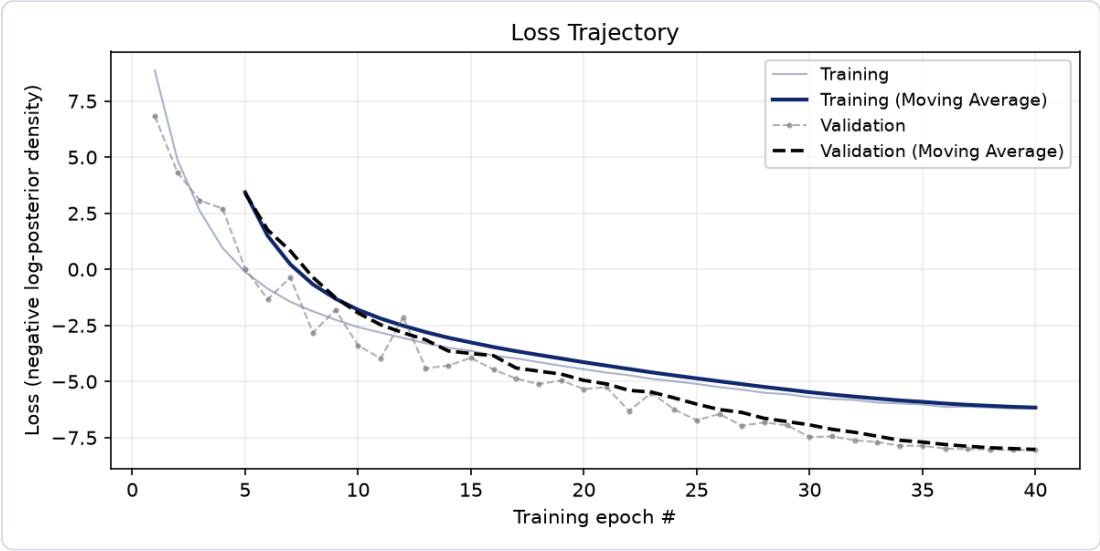


Figure 1. Training and validation loss (negative log-posterior density), 8-parameter model at 50,000 samples. Both decrease monotonically through all 40 epochs, validation staying *below* training the whole way — no overfitting; the early-stopping guard never fired. The earlier 8,000-sample run plateaued at epoch ~17 and then degraded; scaling the dataset eliminated that entirely.

Epoch	Run 6 train	Run 6 validation	Run 7 train	Run 7 validation
1	8.86	6.83	10.25	8.73
10	-2.57	-3.39	-1.17	-2.44
20	-4.45	-5.33	-3.14	-3.21
30	-5.70	-7.47	-4.41	-6.22
40	-6.19	-8.05 (best = final)	-4.93	-6.79 (best = final)

Table 3. Loss milestones. The loss is the negative log-posterior density of the true parameters, so lower is better and negative values are expected once the posterior sharpens beyond the prior. For comparison, the 20,000-sample run ended at -5.87.

6 Results I — parameter recovery

For each of the 300 held-out test systems, 2,000 posterior samples are drawn and the posterior median is compared with the ground truth, with the median absolute deviation (MAD) as the error bar.

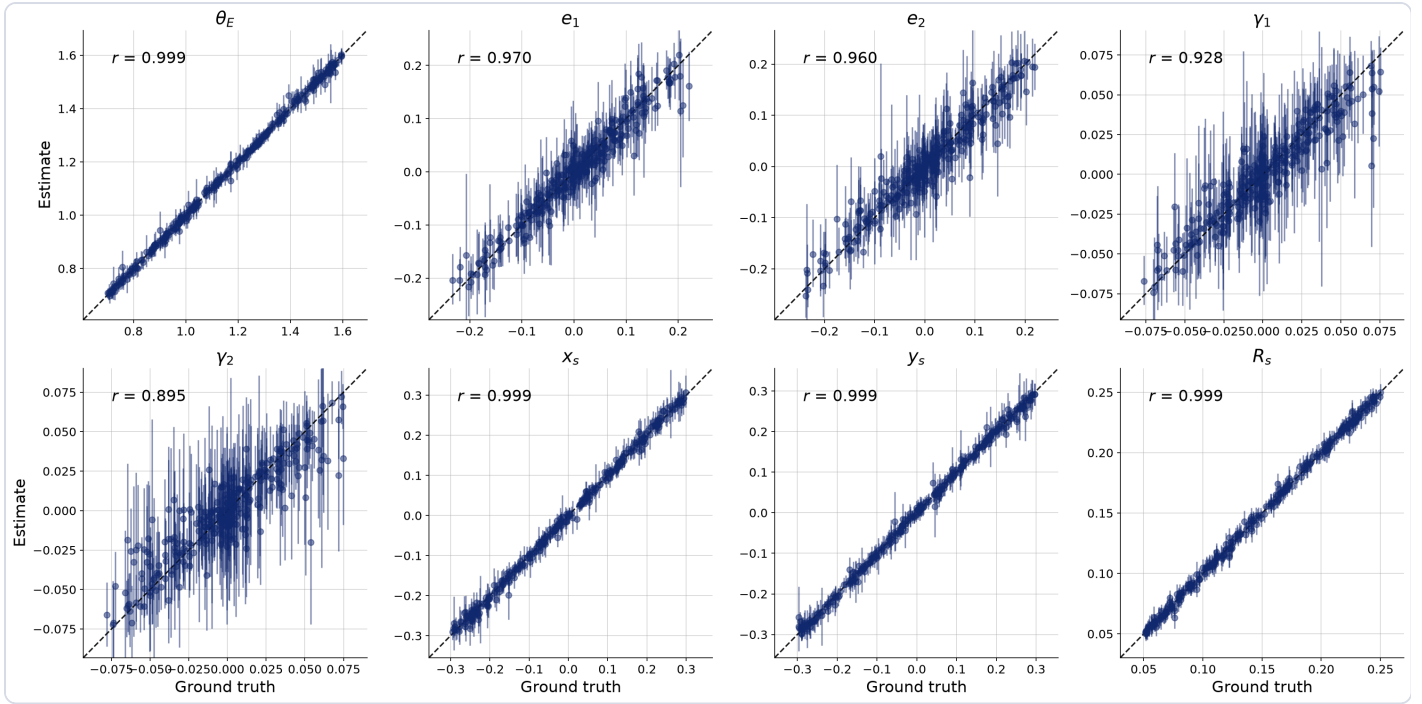


Figure 2. Ground truth vs. posterior median with MAD error bars for all 300 test systems (8-parameter model, 50,000 training samples). Dashed diagonal = perfect recovery; r = Pearson correlation. Note the error bars widening from left to right within each row: the posterior correctly reports shear as its weakest measurement.

Parameter	r (50k)	R^2 (50k)	R^2 (20k)	R^2 (8k, SNR $\approx$ 4)
θ_E	0.999	1.00	1.00	0.88
R_s	0.999	1.00	1.00	0.95
x_s	0.999	1.00	0.99	0.83
y_s	0.999	1.00	0.99	0.84
e_1	0.970	0.94	0.89	0.47
e_2	0.960	0.92	0.84	0.49
Y_1	0.928	0.86	0.80	0.21
Y_2	0.895	0.80	0.65	0.06

Table 4. Recovery accuracy across the three configurations of the same architecture. Raising the SNR (rightmost $\rightarrow$ middle) moved the weakly-imprinting parameters from unrecoverable to good; enlarging the dataset (middle $\rightarrow$ left) sharpened them further.

The recovery hierarchy follows the physics: parameters with large, direct morphological signatures (ring radius, arc position and thickness) are recovered essentially perfectly; the quadrupole imprint of ellipticity is clearly measurable at SNR 16; and even the external shear — whose modulus the prior caps at 0.08 and whose distortion is partially degenerate with ellipticity — is now well constrained.

7 Results II — calibration (SBC) and information gain

Accuracy of the point estimate is not enough; the posterior *width* must also be statistically correct. Simulation-based calibration checks this: across the test set, the rank of the true value within its own posterior samples must be uniformly distributed. We plot the difference between the empirical rank CDF and the uniform CDF, with simultaneous 95% confidence bands.

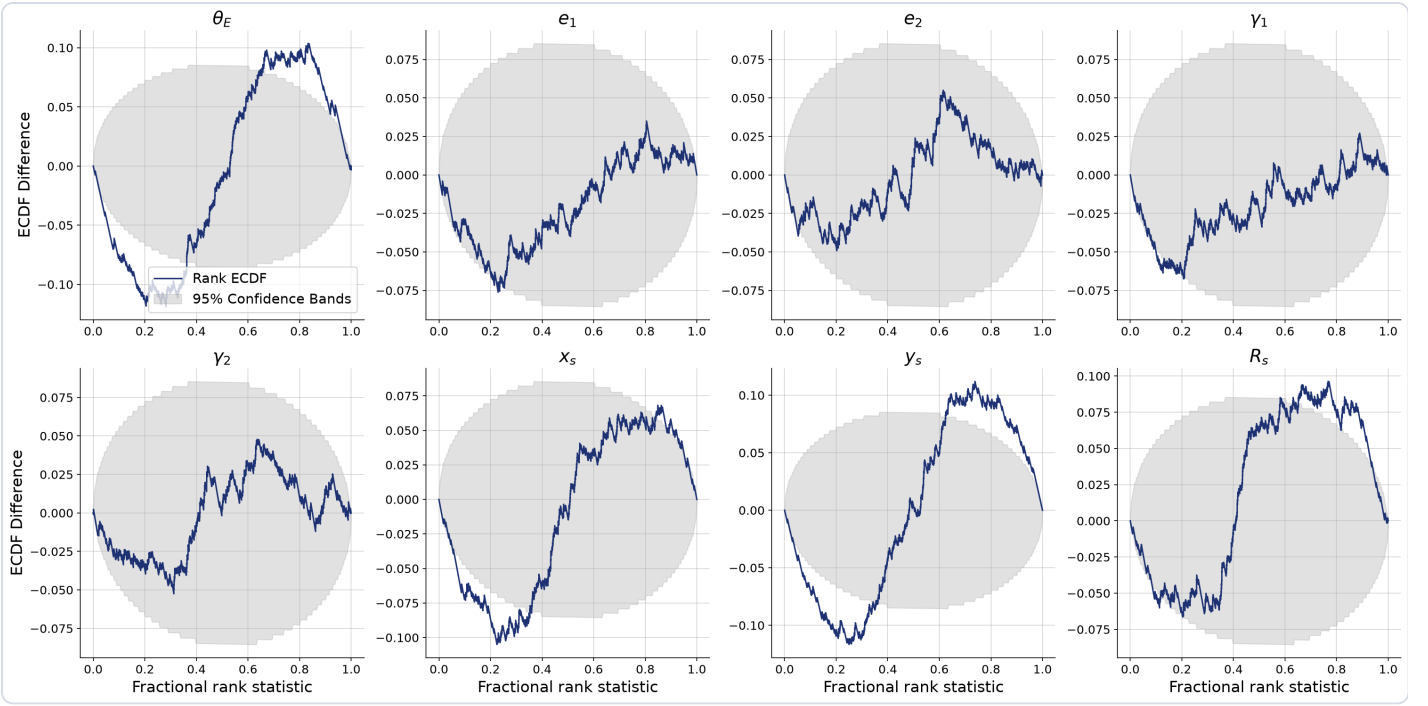


Figure 3. SBC rank-ECDF difference (8-parameter model, 50,000 samples, 300 test systems). Inside the grey band = consistent with calibrated posteriors. $e_1, e_2, \gamma_1, \gamma_2$: calibrated. θ_E, x_s, y_s, R_s : a symmetric S-shape exits the band — too many central ranks — the signature of mildly *underconfident* (too-wide) posteriors.

An honest defect, precisely diagnosed. The mild underconfidence of the four best-measured parameters is the *safe* direction of miscalibration — credible intervals are conservative, never misleading. Crucially, the S-shape is identical at 20,000 and at 50,000 training samples: more data does not touch it. That localizes the cause in network capacity (48-dim summary, depth-4 flow), not data volume — and tells us exactly which knob to turn next.

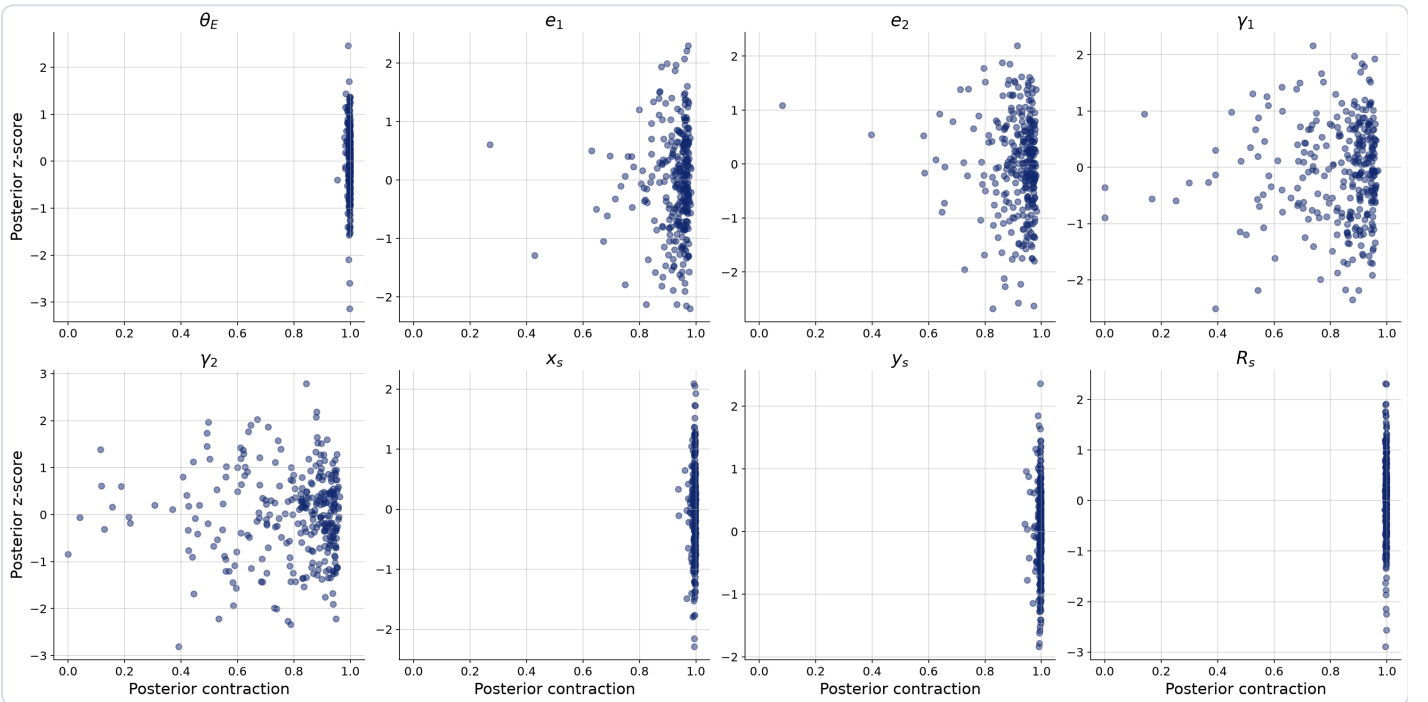


Figure 4. Posterior z-score vs. posterior contraction, one dot per test system. Contraction = $1 - \sigma_{\text{post}}^2 / \sigma_{\text{prior}}^2$ (1 = the image removed all prior uncertainty); z-score = standardized error of the posterior mean. θ_E, x_s, y_s, R_s cluster at contraction ≈ 1 with $|z| \lesssim 2$ — maximal, unbiased information gain; ellipticity and shear reach intermediate contraction with z-scores still centred on zero — less information, honestly reported.

8 Results III — posterior predictive & a complete inference

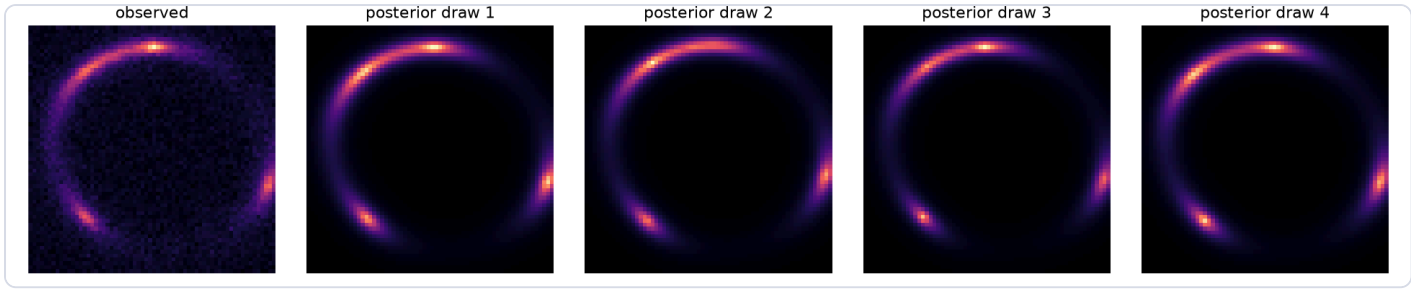


Figure 5. Posterior predictive check. Left: one observed noisy test image. Right: four independent posterior draws pushed back through the simulator (noise-free). Ring radius, knot positions, and the azimuthal light profile are all reproduced; the draws differ mainly in fine knot brightness — exactly the detail the noise genuinely hides.

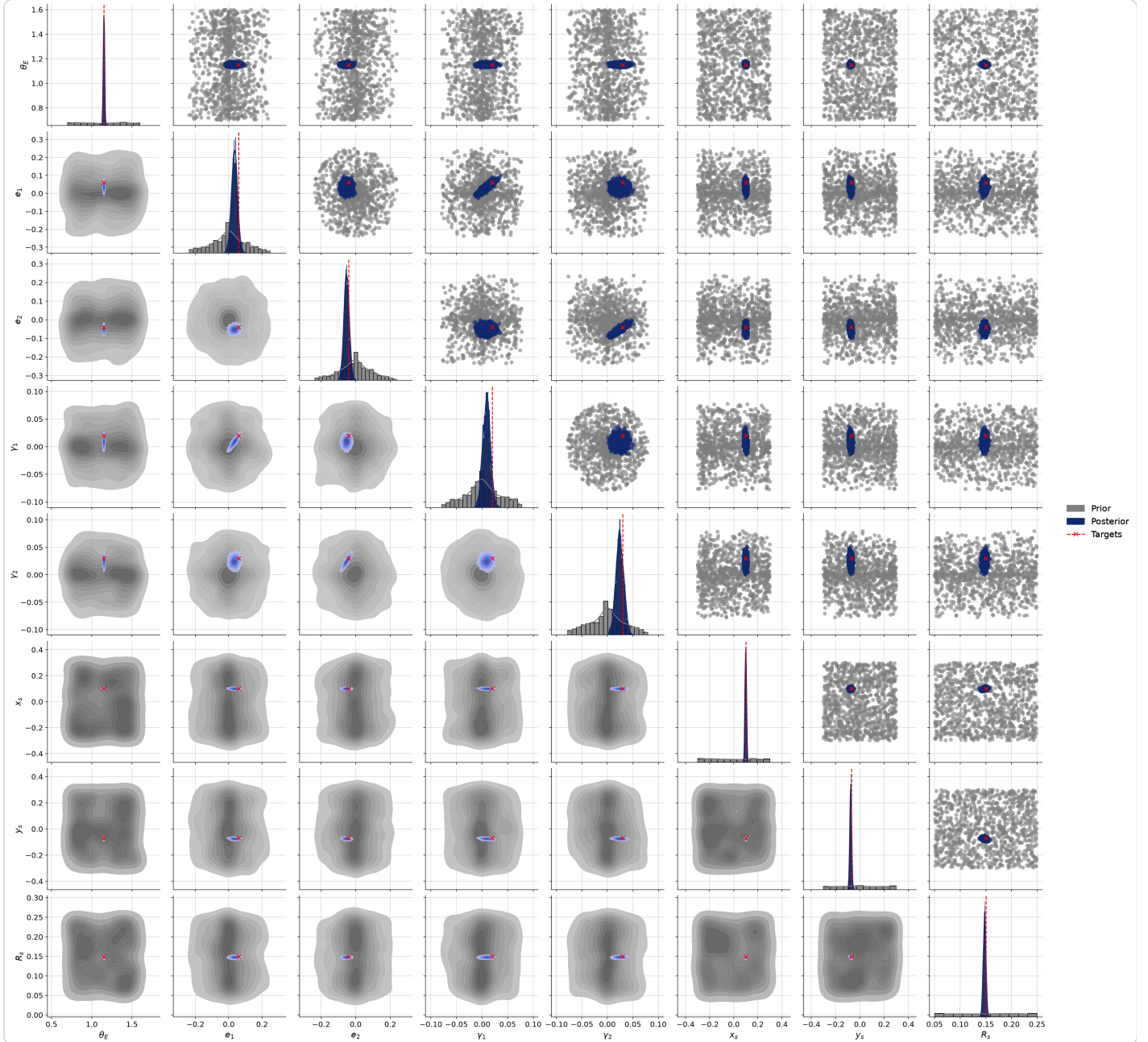


Figure 6. Full inference on one fiducial "observed" system ($\theta_E = 1.15$ ", mildly elliptical lens, weak shear, off-centre compact source): prior (grey), 2,000 posterior draws (blue), truth (red). The posterior contracts dramatically onto the truth for every parameter — $\theta_E = 1.152 \pm 0.008$ against a truth of 1.150 — with the widest remaining spread in the shear components, and visibly tilted joint contours showing the network reports parameter correlations, not just eight separate error bars.

9 The mass-sheet degeneracy (Run 7)

Lensing theory contains a famous blind spot: adding a uniform convergence sheet κ while rescaling the (unobservable) source leaves the image almost unchanged. κ therefore **cannot be measured from a single image** — and a sound inference method must report that impossibility rather than hide it. We repeat the entire pipeline with $\kappa \sim \mathcal{U}(0, 0.2)$ as a ninth parameter, at identical SNR and dataset size.

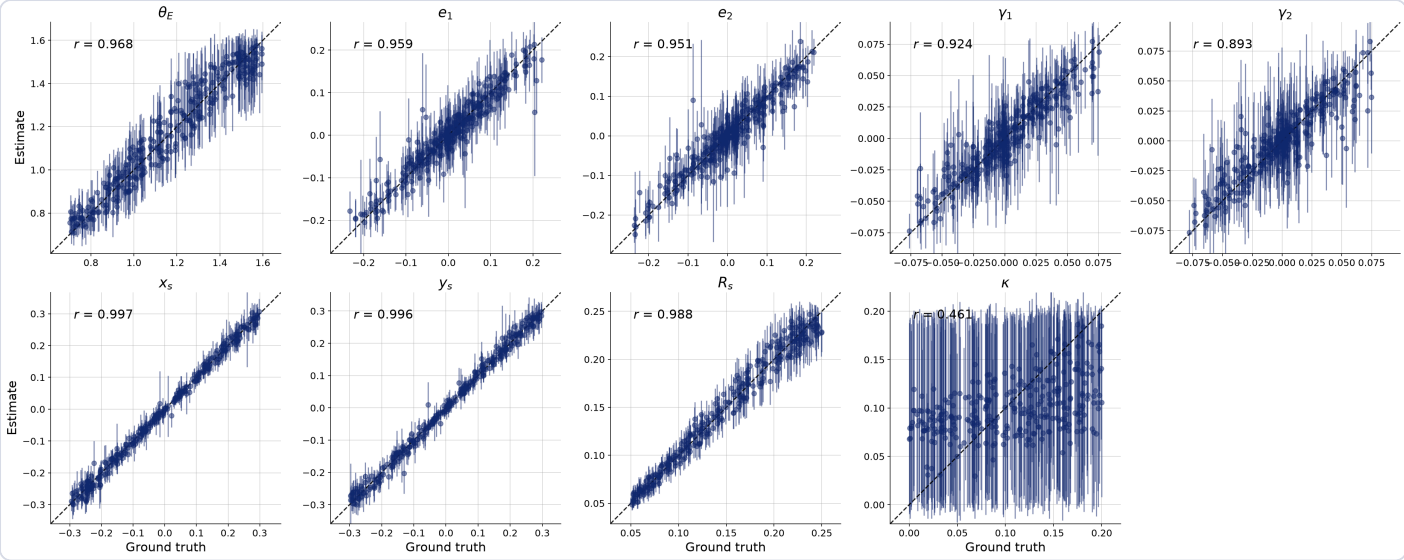


Figure 7. Recovery for the 9-parameter model. The κ panel (bottom right) is a flat cloud with error bars spanning nearly the whole prior — the network says “I don’t know”, exactly as theory demands. The θ_E panel has visibly widened relative to Figure 2; all other panels are unchanged.

Three quantitative signatures, all as predicted:

- **Minimal contraction.** κ prior std 0.058 $\rightarrow$ mean posterior std 0.048: only +16% learned from the image. ($R^2 = 0.21$ — posterior means hover near the prior mean regardless of the truth.)
- **The degeneracy direction is discovered.** Within individual posteriors, $\text{corr}(\kappa, \theta_E) = -0.96$ — the exact trade-off the lens equations predict, found by the network without ever being told. The correlation *sharpened* as data quality improved (-0.83 at SNR 4 $\rightarrow -0.93$ at 20k $\rightarrow -0.96$ at 50k): a sharper posterior hugs the degeneracy ridge more tightly.
- **Surgical damage.** Only θ_E pays: R^2 1.00 $\rightarrow$ 0.94, and in the fiducial-system fit its uncertainty inflates from ± 0.008 to ± 0.065 ($\sim 8\times$). e_1/e_2 : 0.92/0.90; y_1/y_2 : 0.85/0.80; $x_s/y_s/R_s$: 0.99/0.99/0.98 — all unchanged within noise.

The network rediscovered a theorem of gravitational lensing purely from simulated data — the strongest evidence in the project that the learned posteriors are physically meaningful.

10 What changed vs. the earlier runs

Configuration	Overfitting?	Shear recovery	Calibration
8,000 samples, SNR ≈ 4 (Runs 1–3)	Yes — val loss degrades after epoch ~ 17	R^2 0.06–0.21 (unusable)	Overconfident (U-shaped ranks) — the dangerous direction
20,000 samples, SNR ≈ 16 (Runs 4–5)	None	R^2 0.65–0.80	Mild underconfidence for θ_E, x_s, y_s, R_s
50,000 samples, SNR ≈ 16 (Runs 6–7)	None (early stopping never fired)	R^2 0.80–0.86	Same mild underconfidence — unchanged by $2.5\times$ more data

Table 5. The progression cleanly separates the failure modes: SNR determines *which* parameters are learnable at all; dataset size fixed overfitting and sharpened recovery; the residual conservatism responds to neither — it is a capacity limit of the summary/inference networks.

11 Limitations & next steps

- **Calibration slack.** The remaining underconfidence for the best-measured parameters should respond to a larger summary dimension ($48 \rightarrow 96$) or a deeper flow ($4 \rightarrow 6$) — the diagnosed next experiment, not more simulations.

- **Model scope.** The simulator fixes the PSF, lens centroid, source amplitude and profile, and omits the lens galaxy’s own light. Real survey images need these added (and marginalized over); the inference machinery stays identical — that is the point of SBI.
- **No classical benchmark yet.** A per-system lenstronomy MCMC fit on a few test lenses would be a natural cross-check; simulation ground truth is arguably stronger, but the comparison would quantify the speed/accuracy trade directly.
- **Scale.** Once trained, posterior sampling costs milliseconds per lens — the property that matters for the 10^4 – 10^5 lenses expected from Euclid and Rubin-LSST.

12 Reproducing these results

Everything is deterministic given the seed in `src/config.py` (42). On a machine with an NVIDIA GPU:

Step	Command
Environment	<code>python -m venv .venv && pip install -r requirements.txt</code>
Run 6 (8 parameters)	<code>python main.py generate && python main.py train && python main.py evaluate</code>
Run 7 (+ κ)	same three commands with <code>SLI_INCLUDE_KAPPA=1</code> set

All figures land in `figures/`; per-run numbers are logged in `RESULTS.md`. Companion documents: the formal report (`report/report.pdf`), the from-zero beginner’s guide (`report/explanation.pdf`), and the slide deck with speaker notes (`presentation/`).